Section IV: proof repository

Actuator co-design — where the search stops and how far that is from optimal

1 Setup

Hold the link count ℓ and the sensor mask $\mathbf{s}$ fixed. Write $\mathcal{R} := \{1, \dots, N_u\}$ for the actuator index set and, for a retained set $S \subseteq \mathcal{R}$, let $K(S)$ denote $\Pi_\ell(K_\mathrm{d})$ with rows outside S and columns outside $\mathbf{s}$ zeroed, so $K(\mathcal{R}) = K_\mathrm{s}([\ell, \mathbf{1}, \mathbf{s}])$.

$$f(S) := -\, J_\mathrm{LQR}(K(S)) \,/\, J_\mathrm{LQR}(K_\mathrm{d}) \quad \in \; [-\infty,\, 0), \tag{1}$$

$$m_u := \left|\mathrm{supp}\big([\Pi_\ell(K_\mathrm{d})]_{u,:}\big) \cap \mathbf{s}\right|, \tag{2}$$

$$M(S) := \sum_{u \in S} \Big(w_a\, \mathbf{1}\{m_u > 0\} + w_c\, m_u \Big), \tag{3}$$

$$U(S) := f(S) - M(S). \tag{4}$$

Larger f is better; $f = -\infty$ exactly on the destabilizing sets. With ℓ and $\mathbf{s}$ fixed, the sensor and constant terms of J_EA are common to all S, so minimizing J_EA over the actuator mask is equivalent to maximizing U.

Lemma 1 (M is exactly modular)**.** [PROVED] *M is a sum of per-element weights fixed by $(\ell, \mathbf{s})$, hence modular.*

Proof. Distinct rows of $K(S)$ have disjoint supports, so $N_c(K(S)) = \sum_{u \in S} m_u$ exactly. An actuator whose row is emptied by the truncation Π_ℓ carries no controller entries and is not charged w_a by J_EA, which is what the indicator in (3) records. Each summand therefore depends on u alone. □

Remark 2 (why the indicator is not optional). Writing $w_a|S|$ in place of $w_a\,\#\{u \in S : m_u > 0\}$ overcharges every masked-on actuator whose row the truncation emptied. The two coincide whenever ℓ is large enough that every masked row survives, which is why the EA never exposed the difference (ℓ at 90–99% of its maximum) and the greedy baseline immediately did (ℓ at 5–12%): at one greedy fixed point 16 actuators were masked on but only 4 had entries, inflating $M(S^\star)$ from 2.0 to 6.8 and the reported gap from 2.02 to 6.82. All of the curvature of the problem sits in f; by Lemma 1 the structural terms contribute none.

2 Assumptions

Assumption 3 (approximate submodularity)**.** [ASSUMPTION] Let $\mathcal{B}$ be the range of architecture sizes the search occupies. For all S with $|S| \in \mathcal{B}$ and $A + BK(S)$ Schur stable, and all T,

$$\sum_{u \in T} \big[f(S \cup u) - f(S)\big] \;\geq\; \gamma_\mathrm{f}\, \big[f(S \cup T) - f(S)\big], \qquad \gamma_\mathrm{f} > 0. \tag{5}$$

Assumption 4 (monotonicity)**.** [ASSUMPTION] On the same sets, $S \subseteq T \Rightarrow f(S) \leq f(T)$.

Neither is automatic: $K(S)$ is a truncation of K_d, not the optimal gain for the support S.

[measured] The band in Assumption 3 is not cosmetic. Sampling the Das–Kempe quotient (5) stratified by retention fraction, 250 draws per cell, $|T| \geq 2$ (with $|T| = 1$ the quotient is identically 1 and carries no information):

$\lvert S\rvert/N_u$	0.05–0.15	0.15–0.30	0.30–0.45	0.45–0.80
5×5 s1	1.000	0.980	0.981	0.911
5×5 s10	1.000	0.999	0.871	0.879
7×7 s1	0.976	−0.119	−2.012	0.717
7×7 s10	1.000	0.986	0.652	0.110

Entries are min over samples. The searches return $|S^\star|$ of order 10% of N_u, i.e. the leftmost column, where f is submodular. The negative values sit in the middle bands, where the aggregate gain $f(S \cup T) - f(S)$ is near zero and the quotient is dominated by its denominator; 18–26% of draws there are discarded for that reason. Estimating γ_f on the band the search occupies gives $\gamma_\mathrm{f} = 1.0000$ on all three plants of the paper.

[measured] Assumption 4. Over 600 sampled (S, u) pairs with S feasible: 0 violations on the 5×5 grid, 2 on the 7×7, of relative size $\leq 0.8\%$, and no sampled addition destabilized. Script: `scratchpad/probe_A2.m`.

3 Where the search stops

Definition 5 (1-flip local optimum)**.** $\theta^\star$ is a 1-*flip local optimum* if no single actuator or sensor mask flip strictly decreases J_{EA}.

Proposition 6 (elite-child hitting probability)**.** [PROVED] *Let $\mathcal{H}_{t-1}$ be the σ-algebra generated by the population history through generation $t-1$. For every single-bit neighbour θ' of the elite,*

$$\mathbb{P}\big(\theta^{\mathrm{c}} = \theta' \mid \mathcal{H}_{t-1}\big) \;\geq\; q_{\min} \;:=\; \frac{(1-p_c)\, p_m (1-p_m)^{N_u+N_x-1}}{N_p\,(2d+1)} \;>\; 0 \tag{6}$$

uniformly in t, where θ^{c} is some offspring of generation t.

Proof. With probability $1-p_c$ the algorithm skips crossover and copies the fitter of the two selected parents. The elite has the strictly smallest cost, so whenever it is selected it is the fitter one; selection assigns it weight at least $1/N_p$. Mutation then flips exactly the required bit and no other with probability $p_m(1-p_m)^{N_u+N_x-1}$, and leaves ℓ unchanged with probability $1/(2d+1)$. □

Theorem 7 (almost-sure finite-time convergence)**.** [PROVED] *Suppose $J_{\mathrm{EA}}(\theta_0^\star) < \infty$. Then almost surely there is a finite random T with $J_{\mathrm{EA}}(\theta_t^\star) = J_{\mathrm{EA}}(\theta_T^\star)$ for all $t \geq T$, and the limiting gene is a 1-flip local optimum.*

Proof. Θ is finite, so $J_{\mathrm{EA}}(\Theta)$ is a finite value set. Elitism carries the n_e best genes forward unchanged, so $J_{\mathrm{EA}}(\theta_t^\star)$ is non-increasing; a non-increasing sequence in a finite ordered set is eventually constant, at some J_∞. For $t > T$, Proposition 6 gives each prescribed single-bit neighbour probability at least $q_{\min} > 0$ per generation, and $\sum_{t>T} q_{\min} = \infty$, so by the second Borel–Cantelli lemma every such neighbour is produced infinitely often almost surely. If any had cost below J_∞, elitism would carry it forward, contradicting the definition of J_∞. □

Proposition 8 (the returned architecture is irreducible)**.** [PROVED] *At the limiting gene, with $S^\star$ its retained actuator set,*

$$\begin{aligned} u \in S^\star &\implies f(S^\star) - f(S^\star \setminus u) \;\geq\; w_a + w_c m_u, \\ u \notin S^\star &\implies f(S^\star \cup u) - f(S^\star) \;\leq\; w_a + w_c m_u. \end{aligned} \tag{7}$$

Proof. Immediate from Definition 5 applied to $U = f - M$, using Lemma 1 for the marginal of M. □

Each retained actuator pays for itself and each discarded one does not. Both sides of (7) are computable at a cost of $N_u - |S^\star|$ Lyapunov solves, which makes them usable as a stopping test.

Remark 9 (what Theorem 7 does not give). [NOT CLAIMED] No rate. The chain on Θ is irreducible, so the classical result for elitist EAs with full-support mutation would give almost-sure convergence to the global optimum as $t \to \infty$; we do not state it, because it is asymptotic with no rate and holds for *any* elitist GA with full-support mutation.

4 Distance to optimality

The search need not have reached a 1-flip local optimum when the budget expires, so the violation is carried explicitly. For $u \notin S^\star$,

$$\xi_u := \big[\, f(S^\star \cup u) - f(S^\star) - w_a - w_c m_u \,\big]_+, \qquad \Xi(T) := \sum_{u \in T} \xi_u, \tag{8}$$

so that

$$f(S^\star \cup u) - f(S^\star) \;\leq\; w_a + w_c m_u + \xi_u \tag{9}$$

holds with no hypothesis, and $\xi_u = 0$ exactly when u cannot be added profitably.

Theorem 10 (optimality gap)**.** [PROVED UNDER 3, 4] *For any S with $A + BK(S)$ Schur stable,*

$$U(S) - U(S^\star) \;\leq\; M(S^\star \setminus S) \;+\; \big(\gamma_{\mathrm{f}}^{-1} - 1\big)\, M(S \setminus S^\star) \;+\; \gamma_{\mathrm{f}}^{-1}\, \Xi(S \setminus S^\star). \tag{10}$$

Proof. Assumption 3 with $T = S \setminus S^\star$, followed by (9), gives

$$f(S^\star \cup S) - f(S^\star) \;\leq\; \gamma_{\mathrm{f}}^{-1} \sum_{u \in S \setminus S^\star} \big[f(S^\star \cup u) - f(S^\star)\big] \;\leq\; \gamma_{\mathrm{f}}^{-1}\big[M(S \setminus S^\star) + \Xi(S \setminus S^\star)\big].$$

Assumption 4 gives $f(S) \leq f(S^\star \cup S)$. Subtract $M(S)$, split it modularly (Lemma 1) as $M(S) = M(S \setminus S^\star) + M(S \cap S^\star)$, and use $f(S^\star) = U(S^\star) + M(S^\star \setminus S) + M(S^\star \cap S)$. □

Corollary 11 (computable bound)**.** [PROVED UNDER 3, 4] *Since $S^\star \setminus S \subseteq S^\star$ and $S \setminus S^\star \subseteq \mathcal{R} \setminus S^\star$,*

$$\min_{\mathbf{a}} J_{\mathrm{EA}} \;\geq\; J_{\mathrm{EA}}(S^\star) - \mathrm{gap}, \qquad \mathrm{gap} := M(S^\star) + \big(\gamma_{\mathrm{f}}^{-1} - 1\big) M(\mathcal{R} \setminus S^\star) + \gamma_{\mathrm{f}}^{-1}\, \Xi(\mathcal{R} \setminus S^\star), \tag{11}$$

the minimum being over actuator masks at the current $(\ell, \mathbf{s})$. Every term on the right is available from the run.

[measured] Both correction terms vanish in practice. With $\gamma_{\mathrm{f}} = 1$ on the operating band, (11) collapses to

$$\text{gap} \;=\; M(S^\star) \;+\; \Xi(\mathcal{R} \setminus S^\star) \tag{12}$$

— the structural price of what was kept, plus what was left on the table by not converging.

plant, seed	γ_{f}	gap	$J_{\mathrm{EA}}(S^\star)$	$\min_{\mathbf{a}} J_{\mathrm{EA}} \geq$	Thm. 10 hypothesis
5×5, 1	1.0000	2.100	6.775	4.675	150/150, end ✓
5×5, 10	1.0000	1.450	7.811	6.361	63/150, end ✓
5×5, 15	1.0000	0.975	6.547	5.572	24/150, end ×
7×7, 1	1.0000	2.200	7.646	5.446	144/150, end ✓
7×7, 10	1.0000	2.248	10.437	8.189	95/150, end ×
7×7, 15	1.0000	7.199	11.185	3.986	22/150, end ×
IEEE 13-bus	1.0000	0.450	2.448	1.998	149/150, end ✓

Where $\Xi = 0$ the gap *is* $M(S^\star)$: 18–31% of J_{EA}. The outlier, 7×7 seed 15 at 64%, is slack-dominated ($\Xi \approx 5.2$), and that seed is 1-flip locally optimal in only 22 of 150 generations. An unfinished run receives a wider band, not a tighter unjustified one.

Remark 12 (architecture size is set by the weights). [PROVED] Every term of J_{EA} is non-negative, so

$$w_a\, N_a(S^\star) \;\leq\; J_{\mathrm{EA}}(S^\star) \qquad \Longrightarrow \qquad N_a(S^\star) \;\leq\; J_{\mathrm{EA}}(S^\star)/w_a. \tag{13}$$

The LQ term is normalized to 1 at K_{d}, so the entire LQ budget available to pay for actuators is $O(1)$ while each costs w_a: the actuator count at the optimum is set by the weights, not by the system size. Measured, $J_{\mathrm{EA}}(S^\star)/w_a$ is 13–28 against actual $|S^\star|$ of 2–7.

Remark 13 (scope). Three limits. *(i)* (11) bounds the optimum over actuator masks at the elite's current ℓ and $\mathbf{s}$; relaxing those two coordinates can buy more than the entire certified gap, so the band is not a bound on the global optimum. Measured, the greedy architecture costs less than the bound on 2 of 6 runs — no contradiction, since greedy varies ℓ and $\mathbf{s}$. The restriction is forced: f restricted to sensor sets is $-\infty$ on most subsets and admits no submodularity ratio. *(ii)* Theorem 10 and Proposition 8 are statements about a *point*, not an algorithm; they hold for any method returning a 1-flip local optimum. Only Proposition 6, the proof of Theorem 7, and Remark 14 are specific to the EA. *(iii)* γ_{f} and Assumption 4 are estimated, not proved, so the results are rigorous conditional on constants we measure but cannot certify.

Remark 14 (mutation-rate scaling). [PROVED + MEASURED] The factor $(1 - p_m)^{N_u + N_x - 1}$ in (6) decays exponentially in the gene length at fixed p_m, but $p_m = c/(N_u + N_x)$ gives $(1 - p_m)^{N_u + N_x} \to e^{-c}$, so $q_{\min}$ becomes dimension-free. Measured, $(1 - 1/n)^n = 0.3654$ at $n = 75$ and 0.3666 at $n = 147$, against 0.0213 and 5.31×10^{-4} for fixed $p_m = 0.05$. Ten-seed paired test of $p_m = 1/n$ against $p_m = 0.05$: +1.56% on the 5×5 grid (5/10 wins, $p = 0.183$), +11.36% on the 7×7 (9/10 wins, $t = 3.37$, $p = 0.0083$). The predicted signature is not "scaled is better" but "the advantage grows with n", and that is what appears.

5 The greedy baseline

Greedy round-robin local search over $\{\ell, \mathbf{a}, \mathbf{s}\}$, each phase run to a fixed point, with the link phase restricted to the EA's own $\pm d$ neighbourhood.

Proposition 15 (the descent path never leaves the stable set)**.** [PROVED] *Starting from K_{d}, which stabilizes by construction, and accepting only strict improvements, every iterate is feasible.*

Proof. $J_{\mathrm{EA}} = +\infty$ on infeasible points, so any infeasible candidate has cost strictly greater than the current finite incumbent and is never accepted. □

[MEASURED] Feasible on 6/6 runs, while 4–10 infeasible candidates are probed and rejected (215–251 on the open-loop unstable plant). The EA has no such property: mutation samples infeasible genes unconditionally, and it probes 189–332 of them per run. This contradicts the argument that removing a link risks "an infinite jump greedy cannot back out of": on these plants one never has to pass through the infeasible set at all.

Proposition 16 (Theorem 10 holds for greedy in its tight form)**.** [PROVED] *Greedy's stopping condition — no single-bit flip improves J_{EA} — is strictly stronger than the addition half used in* (9)*, so $\xi_u \equiv 0$ and* (11) *reduces to $M(S^\star) + (\gamma_{\mathrm{f}}^{-1} - 1)M(\mathcal{R} \setminus S^\star)$ by construction, irrespective of convergence.*

[MEASURED] Evaluated at the greedy fixed point the gap is 23–31%, against 15–64% at the EA's, whose spread is driven by Ξ.

Proposition 17 (evaluation count)**.** [ASSUMPTION + PROVED] *If no masked-off element is ever re-added, each element is removed at most once, so the accepted mask moves number at most $N_u + N_x$. The actuator phase costs N_u evaluations per sweep and either accepts a move or ends the phase, and likewise for sensors, so with C cycles*

$$\textit{evaluations} \;\le\; N_u(A + C) + N_x(S + C) + (\textit{link phase}) \;=\; O\big((N_u + N_x)^2\big). \tag{14}$$

[MEASURED] Re-additions: 0 on all seven runs. Sensor removals: 0 on all seven — the sensor mask is never touched, and the sparsity in N_s is induced by the actuator mask and ℓ. Cycles: exactly 2 everywhere, which follows from the sensor phase accepting nothing and so never perturbing the actuator fixed point. Evaluations/$(N_u + N_x)^2$: 0.077–0.147, decreasing in n, so (14) is the right shape but loose — accepted moves grow much more slowly than n.

[NOT CLAIMED] *Absence of re-additions is empirical, not a consequence of Assumption 3.* Submodularity means diminishing returns, so after removing other elements the marginal value of u can only *increase*, which would make re-adding it more attractive, not less.

6 What is retired from the submitted Section IV

Submitted result	Reason for removal
$\Phi(h)$, $h^\star$, Lemma 3	certifies 0/150 generations; $h_t \equiv 1 < h^\star$
Definition 2(c) (h_t)	hop distance is a proxy for a magnitude-ordered prune
Lemma 4	needs K_{t-1} entrywise equal to K_{d} on its support
Propositions 1, 2	consequences of Lemmas 3, 4; crossover is uniform, not single-point
Theorem 1 (drift)	single-path counting is 10^2–10^4 loose; see below
$\mathbf{a} = \mathbf{s} = \mathbf{1}$	assumes away the coordinate that does the work

[MEASURED] The drift route cannot be rescued by a better certificate: substituting the *exact* LQ increase for the cost bound — the ceiling for any certificate — still leaves it 17.8× (5×5) and 196× (7×7) below the measured decrease. The residual sits in the probability factor $(1 - p_m)^{n-c}$, which is forced: for the elite child the flipped-bit set must contain no bit outside the certified set, because $J_{\mathrm{EA}} = +\infty$ on unstable genes and the harm of an arbitrary flip therefore has no bound.